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Swarm AIGC Agent Based Approach for Shale Oil Wellbore Operation Policy Decision Supporting: Field Implementation in Western China

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Abstract

Shale-oilplayseven significant role in ensuring energy supply and optimising the energy consumption in China. Withcontinuous increment ofshale-oil demanding, even high restrictions are shifted towards wellbore reliability and maintenance including production and quality relied planning execution, early warning, operational optimisation, and integration of technology, business and finance. However, reliability management and executing ofwellbores downhole operation plans, including problems solving of pump leakage, rod breakage, eccentric wear, wax deposition, as well as workover planning, corresponding work plan formulation, spare parts management, and financial assessment, still strongly rely on experiences and manual efforts. With continuous increments of assets reliability and production efficiency, this leaves significant room for improvement in work efficiency, plan execution consistency, inventory turnover rate, as well as financial and economic efficiency.

This manuscript proposes a swarm AIGC agent approach (SAA) for autonomous wellbore downhole operation and continuous optimising based on latent space simulation assisting. Through innovative architectural design, the SAA integrates the process intelligence and logical reasoning advantages of intelligent agents, stimulating the deep thinking and execution capabilities of the foundational model. By combining expert experiences and literature data coming from China's shale oilfields, an mixture of experts (MOE) system is conducted to achieve problem identification, swarm agents assemble, autonomous task flow orchestration and execution, and decision support for downhole operation strategies driven by financial and business insights. Simulation of hidden space evolution (HSD) is generated and taken into account for aiding the further analyses of wellbore deterioration. Multiple modality of wellbore downhole operation datafrom a major shale-oil field allocated western China were used as stimulating data for fine tuning, the well-trained system validates the performance in variousactual application scenes, and comparison were made between classical and SAA approach.

Actual statisticsverified that the proposed approach demonstrates good performance in decisionmakingscenarios including abnormal spotting, solutions generation and recommendation, code

generating, agents generating and orchestration, workflow creation and execution, outcomes evaluation, refinements, and further recommendations among material planning and financial supporting, with an overall range of cosine similarity varied between 0.78 and 0.86. Thisactivates the SAA based approaches to engage the decision supportupon the whole workflow of shale-oil acquisition, enabling potential autonomous methodology for lowering currently complex task-chainof downhole operating through an in-depth business-finance integrating.

This manuscriptverified that the proposed SAA approach is suitable for automatically and autonomously dealing with the complex downhole implementation workflow. Utilising the proposed SAAto form a hidden space, the success case provide a potential innovative decision support scheme for continuous refined, fullvaluechain decision-making and optimal business-finance collaboration in the context of less fault tolerant, high efficiency required and unseenscenarios. The generalisable methods and standardised products designed under this practice will be beneficial for further achieving generalisability and process optimisation of complex wellbore operation workflows in the future.

Key words: Shale oil, Wellbore downhole operating, Swarm agents, Latent space simulation, Autonomous workflow orchestration and execution

Introduction

China's shale oil development has now emerged as a critical component of national energy supply strategies, with major basins allocatedwestern of country play main roles of shale oil contributions (Wang et al. 2024). Despite significant progress in drilling and multi-stage fracturing innovation, wellbore downhole operational efficiency in China's shale oilfields moved forward significantly. However,shale oil producer faces systematic challenges rooted in conventional management paradigms. Specifically, current practices predominantly rely on humandriven decisionmaking for wellbore fault diagnosis, operational adjustments, and abnormal triggering; the maintenance often supplemented by subcontracted maintenance teams to balance economic constraints and safety requirements. This model might influencing the cumulative non-productive time (NPT), wising gaps between centralised monitoring and decentralised field execution, as well as brings systematic finance-operational inefficiencies:

1. Fragmented fault response mechanisms leading to increments percentages of NPT.
2. Suboptimal resource allocation driven inconsistency regarding to centralised decision and distributed execution.
3. Knowledge silos between technical experts and frontline operators might also cause inconsistency between decision and execution.
4. Reactive based maintenance protocols triggered by abnormal signatures or production declines cause delay of just-in-time maintenance, and economic trade-off becomes particularly acute in marginal conditions.
5. Traditional information systems, though effective in data archiving and process management, strongly rely on human capacity and lack of seamless.

Taking a major shale oil producer in western China as an example, from 2021 to 2024, a total of 2,160 wellbore downhole operations were conducted, including 1,975 routine pump inspections, 185 electrical submersible pump (ESP) inspections, 235 well stimulation/treatment operations, and 11 major maintenance. The cumulative downtime exceeded 40,000 hours, effected approximately 27,000 tonnes of oil. These limitations in wellbore operation present significant opportunities for improvement. Given the current high reliability of wellbores, abnormal operating conditions are even rare Consequently, in practical field operations, well condition assessments and remedial workover strategies are based solely on binary labels of ‘normal' or ‘abnormal' operating conditions. If the intermediate states of abnormal wellbore condition progression and the actual dynamic processes of wellboredeterioration could be identified, it would be feasible to develop lower-cost wellbore downhole operation policies in advance. This would also provide auxiliary decision-making data support for spare parts inventory planning, production forecasting, operational scheduling, as well as other operational activities.

Unlike traditional discriminative based AI, which is confined to single-use scenarios and operates by making judgements based solely on pre-existing data, generative AI (GenAI) introduces transformative human-AI interaction paradigms. By substituting logical determination with content generation, GenAI enables AI systems to simultaneously perform applications, engage in associative reasoning, and act as connectors to integrate disparate standalone applications, thereby facilitating the resolution of complex multi-step tasks (Chen et al. 2024). However, the direct application and reasoning processes of GenAI are constrained by several limitations, including challenges in learning proprietary industrial corpora, inadequate comprehension of end-users' domain-specific intentions, and insufficient design of human-AI interaction frameworks relying on prompt engineering, chain-of-thought reasoning, as well as SFT etc (Chen et al.2025). To address these gaps, the integration of an agent-based framework is necessary to be taken into account to refine GenAI prompt engineering and fully unlock its potentials. Thanks to the impressive transition of generative AI and corresponding agents architecture on key human-machine interaction evolution, including human-machine dialogue with deep thinking (Deepseek. 2025), content generation (OpenAI. 2025), professional knowledge retrieval (Edge, D et al. 2024), complex logical thinking (Schmidgall et al. 2025), task execution (Feng et al. 2025), and reflection and refinement(Ocker et al. 2025), it has become possible to achieve explainable human-machine interactive based decision-supporting under event triggering scenarios.However, in the context of the aforementioned scenarios involving classification, discrimination, prediction and execution, the majority of applied research has predominantly focused on generating estimates of future states based on existing data and knowledge, with comparatively limited exploration of the dynamic evolutionary processes underlying these three categories of scenarios.

Currently, in typical upstreamenergy industries such as shale oil production and operations, the following key challenges persist: Despite the maturation of downhole operational techniques and the sustained improvement in the reliability of high-value asset like wellbores, naturally occurring abnormal operating condition data remain scarce, making it difficult to meet users' demands for optimising downhole operational strategies based on the full life cycle of wellbore conditions. Users seek to leverage multi-dimensional data to formulate production plans guided by economic performance evaluations. They aim to describe the evolutionary states of critical equipment or value assets through the generation and analysis of intermediate-state data, thereby informing maintenance scheduling and spare parts inventory management. Furthermore, users aspire to achieve autonomous management loops by: (1) identify issues through reasoning chains, (2) generate solutions, (3) execute tasks, and (4) reflect on the rationality of outcomes. Through the reinforcement of generative AI foundation models by agents, autonomous problem identification, data generation for evolutionary processes (i.e. HSD), task execution, and optimisation are possible to be realised without extensive fine-tuning but just using agents to stimulate it.

Addressing the aforementioned challenges of scientific/technical problem discovery, task decomposition, and executing workflow generation within latent spaces (Ren et al. 2025), this manuscript proposes the SAA systemdesigned to generate descriptive data capturing process evolution, which is currently lacking in decision support scenarios such as prediction, analysis, autonomous task executing. Drawing inspiration from aswarm agent based attention spreading mechanism (Chen et al. 2025) and integrating a generative agent architecture, the SAA system constructs a swarm agent architecture comprising four roles and multiple distinct functions, each responsible for executing a closed-loop workflow involving issue finding, solution formulation, execution, rationalisation reflection, feedback-driven optimisation, and iterative refinement. Within each individual agent, a self-supervised agent mechanism is employed to implement self-supervised feedback loops, thereby optimising decision-making behaviours through autonomous agent refinement. This approach substitutes traditional fine-tuning, significantly reducing both the time and computational overhead associated with application development while lowering the barriers to deploying intelligent capabilities. The proposed methodology proposes a paradigm shift through the deployment of a GenAI based swarm agent approach specifically engineered for shale oil wellbore downhole cognitive operation. Building upon one of major oil field infrastructure, the system integrates eight specialised agents functioning as collaborative decision-making and autonomousallies process, namely, (1) anomaly detection/ predictionagent (ADA), (2) fault analytics agent (FAA), (3) maintenance policy agent (MA), (4) repairing policy recommendation agent (RPRA), (5) solution generation agent (RGA), (6) execution orchestration agent (EOA), (7) financial evaluation agent (FEA), (8) refinery and knowledge accumulate agent (RKA), with sub-agents embedded to form the distributed agents, thereby achieving full cognitive processing and executing to enrich the operational efficiency regarding to wellbore downhole operation.

The remaining parts of manuscript are organised as follow: section3 delivers the concept of GenAI based hidden space to form the intermediate conditions between health and failure of wellbore integrity, section 4 proposes the architecture of SAA combined with hidden space data generation for describing deterioration of wellbore, thereby supporting the downhole maintenance policies, experimental results and discussions are given in section 5, for analysing approach performance of decision supporting among complex wellbore downhole operation of shale oil producing, by aiding the HSD simulation data; concluding remarks is given in section 6.

Hidden space to form the development of intermediate states

In current energy sectors, according to the maturation of technological processes and operational conditions, as well as significant improvements in asset reliability, abnormal operating conditions have become low probability events. Through sensing networks and condition monitoring techniques, the data collected on physical changes in monitored objects predominantly fall into two categories: healthy (H) and failure (F) states. For a given subject θ, the relationship between the healthy state (H_θ) and the failure state (F_θ) can be expressed by equation (1):

$$H_\theta \rightarrow F_\theta \tag{1}$$

Moreover, the continual enhancement of reliability has rendered it challenging to acquire intermediate state data that describe subtle evolutionary processes—data residing in the latent space between healthy and failed states, referred to asHSD. Leveraging the capabilities of generative AI agents for complex task decomposition, semantic understanding, structural description, enhanced retrieval, and associative generation, this study investigates a method to generate intermediatestate evolutionary data within the latent space, as illustrated by equation (2) using generative agents(Wang et al., 2024). This approach aims to produce intermediatestate simulation data at a lower cost, thereby better supporting scenarios involving process evolution, predictive reasoning, and decision supporting. Additionally, it facilitates decisionmaking activities in operational domains, such as planning, supply chain preparation, the formulation of productions, material planning, etc.

$$H_\theta \rightarrow H_{\mathrm{i}\theta} \rightarrow F_\theta \tag{2}$$

SAA architecture

The SAA system employs an autonomous construction technique for generative agent clusters based on an attention spreading mechanism previously published (Chen et al. 2025), integrating online query information with industryspecific knowledge graphs to form MOE system to enable problem identification, task decomposition, workflow generation, and task execution within latent spaces, while facilitating conclusion derivation and feedback. In terms of system architecture, as illustrated in Fig. 1, the system comprises four modular components: (1) novel idea exploration and generation, (2) task decomposition and plan formulation, (3) workflow decomposition and execution, as well as (4) summarisation and feedback. Each module establishes rolespecific distributed agent clusters through the agent cluster construction and attention spreading mechanism described by (Chen et al. 2024), augmenting promptbased thought chains to simulate human cognitive logic. This approach supports complex scientific/technical problem refinement and technical route development, including scientific/technical question generation, solution conceptualisation, task decomposition and planning, workflow generation and correction, workflow execution, and summarisation/feedback.

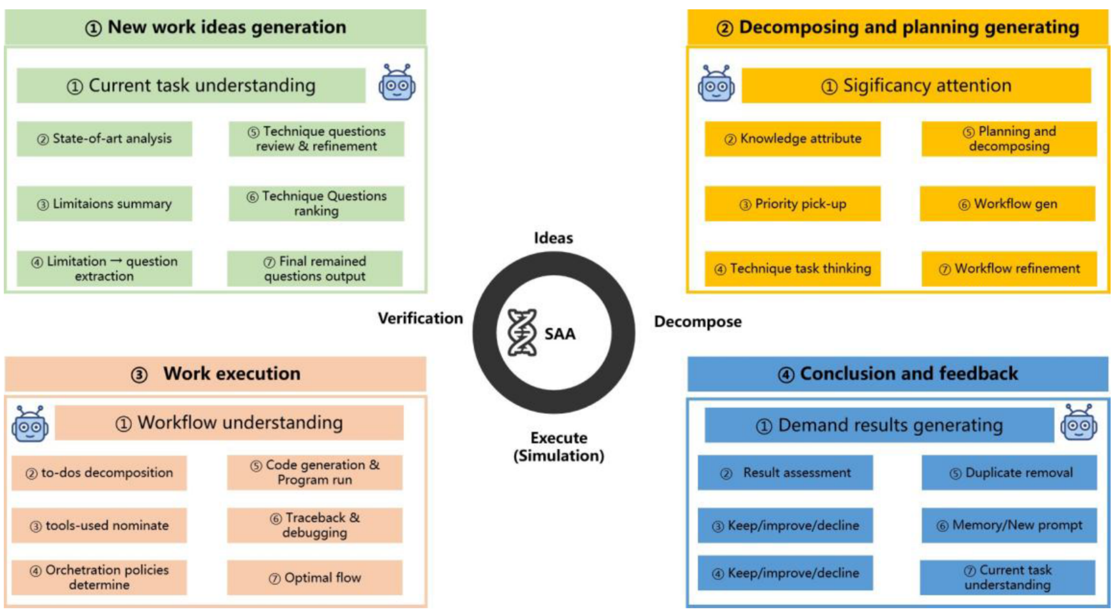


Figure 1—Concept of SAA architecture

In terms of foundation model implementation, the system leverages multiple AIGC foundational models of varying scales to drive agentbased gametheoretic interactions (Chen et al. 2025). Focusing on intermediatestate search, simulation, and inference in latent spaces, it executes functions such as technical question generation, task decomposition, plan formulation, task execution, and feedback analysis. During the generation phase, an LLM assists in auxiliary retrieval and associative reasoning through online queries of historical data, public literature, and pre-constructed knowledge graphs (Wang et al. 2024). A dedicated generative agent formulates technical questions, while a verification agent (Xu et al. 2024) ensures quality control, forming a closedloop system. Validated technical questions undergo solution generation by the verification agent, with intermediatestate problems prioritised and validated via designated thought chains. Continuous refinery process is employed until inter-agent consistency exceeds 90%.

In terms of complex task decomposition and execution, generated technical questions and technical routes are decomposed into executable plans through a structured workflow. Dual generative agents perform internal validation, achieving ≥90% consistency through iterative refinement. Problem analysis and content decomposition follow analogous validation procedures. In terms of latent space data generation, using decomposed latent space tasks and execution plans, the system generates code to interface with mathematical simulation or industrial software tools, producing synthetic data. The agents initially process these data, filtering invalid cases (bad cases) and returning them to the prompt set. Subsequent PCA coding (MATLAB/Python) projects datasets into two/three-dimensional spaces, with calibration against historical positive/negative examples. Evolutionary trajectory compliance is assessed using multivariate based clustering (Xiong et al., 2024), enabling secondary data validation.

For the refinery process, sub-agents are selected based on pattern classification requirements. Through agent-based voting mechanisms, agent decomposes classification tasks by retrieving user intent from prompt sets, combining latent space data interpretation to generate task-specific code. This iteration selects appropriate algorithms, executing tasks from data preprocessing to performance evaluation. For result optimisation, a verification agent group reviews the entire generation workflow and outcomes, formulating optimisation suggestionsbased on reasoning. Iterative refinement continues until the process meets user demands.

SAA based multi-role agents for supporting wellbore downhole operation

The system identifies and prioritises multiple potential technical issues through distributed generative agents, optimally sequencing them for resolution (Fig. 2). The ADA transmits processed results and decomposed task steps to FAA. The FAA refines the likelihood rankings of ADAgenerated outcomes by integrating ADA's analytical interpretations, historical resolution records, and online query conclusions, then forwards the most probable execution plan to MA. With MA generates optimal well maintenance measures based on FAA-transmitted data, which are subsequently communicated to the RPRA and RGA. These agents respectively recommend wellbore maintenance strategies and produce detailed wellbore repair plans. Upon completion of both wellbore maintenance strategies and repair plans, the agent cluster interfaces with FEA systems, leveraging the SAA's cross-domain knowledge retrieval and generative capabilities to execute equations (1) and (2), thereby forming HSD for intermediate based decision supporting. Through latent space cross-simulations between operational and finance, the system provides auxiliary optimisation recommendations for well maintenance strategies and plans from a financial evaluation perspective, ensuring precise outcomes.

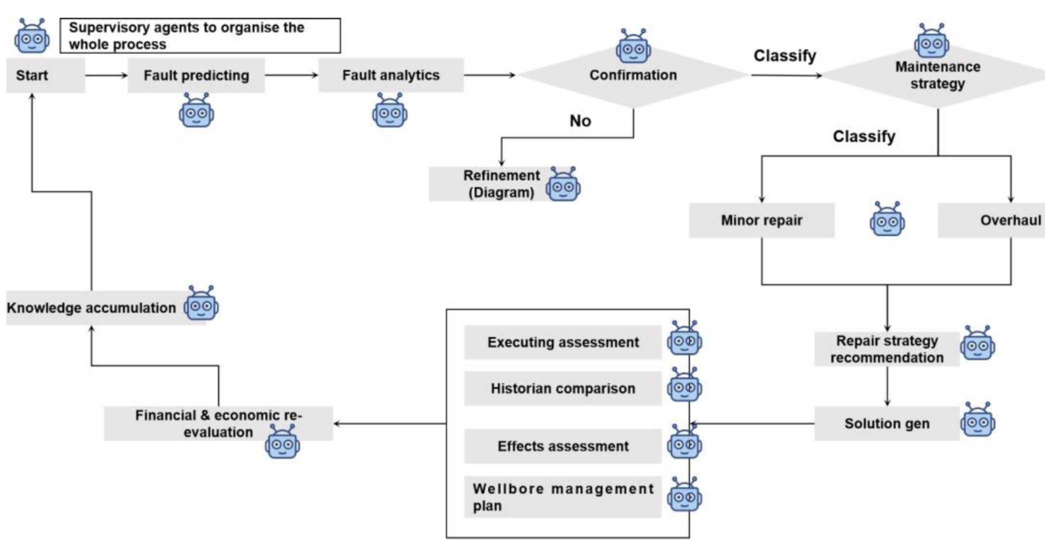


Figure 2—Operational workflow of wellbore downhole operation decision support aided by SAA

To enable operational efficiency, the SAA first retrieves historical resolution protocols and effectiveness metrics when handling abnormal events. If historical precedents exist, HSD solutions are generated based on recorded data. For unprecedented faults, the system conducts online literature reviews combined with HIL assistance and latent space imagination techniques to develop intermediatestate solutions and corresponding well maintenance plans. After selecting a definitive repair approach, the agent cluster generates fault analytic and maintenance codes,for workflow execution scripts, functional diagram classification, well anomaly solutions derived from functional diagrams, and material requirement plans, respectively.

These outputs are then assigned to appropriate operational teams for implementation. Following well maintenance completion and result reporting, the task is formally closed, with outcomes processed through the RKA to reinforce the SAA's refinement. This continuous feedback mechanism enhances the performance of SAA.

Within the SAA, the distributed generative agent cluster employs an attention spreading mechanism (Chen et al. 2025) to facilitate divergent and convergent thinking processes, thereby identifying the most rational problem sets. Given that latent space evolution and task decomposition constitute composite task chain resolution under multiple constraints, their outputs manifest as fuzzy mapping relationships among multi-dimensional input variables.

To address this SAA approach, precise user intent localisation and controllable data generation processes are critical for task realisation. Based on multiple rounds of preliminary experiments, a prompting tips along with general workflow tailored for shale oil wellbore downhole operation systems has been squeezed, with detailed procedural specifications are provided in algorithm 1:

Algorithm 1—General workflow and prompting tips for SAA processing

1. Role Declaration

The agent's name is set as "YourAgentName". Its role is to oversee the entire process, from identifying problems to executing tasks and refining the workflow within a closed - loop system.

2. Task Understanding

2.1 Current Task Comprehension

This function extracts the objective, scope, and constraints from the task description. It returns an array containing these three elements for further processing.

2.2 State - of - art Analysis

It conducts research on the given task domain and returns the research results, which help in understanding the existing solutions and technologies related to the task.

2.3 Limitations Summary

Identifies all the limitations within the task information and returns them. This is crucial for planning and avoiding potential issues during task execution.

3. Agent Role Allocation

Allocates roles to sub - agents based on the task parts. It creates a dictionary where the keys are sub - agents and the values are the corresponding task parts they are assigned to.

4. Thought Chain Understanding

4.1 Technique Questions Review and Refinement

First, generates initial technique - related questions for the task domain. Then, refines these questions according to the specific context of the task, resulting in more targeted and useful questions.

4.2 Technique Questions Ranking

Ranks the refined questions based on certain criteria (such as importance or impact) and returns them in an ordered list.

5. Task Decomposition Strategy

5.1 Significance Attention

Identifies the most critical aspects of the task information. This helps in focusing efforts on the important parts of the task during decomposition and execution.

5.2 Knowledge Attribute

Classifies the knowledge required for each subtask. Understanding the knowledge needs is essential for proper resource allocation and task execution.

5.3 Priority Pick - up

Determines the order in which subtasks should be executed. This takes into account dependencies between subtasks and their relative importance to the overall task.

5.4 Planning and Decomposing

Breaks down the main task into smaller, more manageable subtasks. This makes the task easier to handle and execute.

5.5 Workflow Generation

Creates a workflow based on the subtasks. The workflow defines the sequence and relationships between the subtasks, guiding the task execution process.

5.6 Workflow Refinement

Updates the existing workflow using new information. This ensures that the workflow remains relevant and efficient as the task progresses.

6. Task Execution Strategy

6.1 Workflow Understanding

Checks whether the agent has a clear understanding of the workflow. If the agent can understand the workflow, it returns true; otherwise, it returns false, indicating that further clarification may be needed.

6.2 To - dos Decomposition

Further breaks down each subtask into individual to - do items. These to - do items are the actionable steps for task execution.

6.3 Tools - used Nominate

Identifies the tools required for each to - do item. Selecting the right tools is important for the successful completion of tasks.

6.4 Orchestration Policies Determine

Sets the policies for coordinating different task parts. This is especially important when multiple agents or teams are involved in the task execution.

6.5 Code Generation and Program Run

If the task requires programming, it generates the code and runs the program. It returns the result of the program run, or null if the task does not involve programming.

6.6 Traceback and Debugging

Tries to fix the issues by tracing back the steps and debugging. It returns the fixed issues, helping to ensure the smooth running of the task.

6.7 Optimal Flow

Optimises the execution process to improve efficiency. It returns the optimised execution process, which can save time and resources.

7. Feedback Optimization Strategy

7.1 Generate Demand Results

Generates the results according to the task requirements and returns them. These results are the output of the task execution.

7.2 Result Assessment

Evaluates the generated results against the task goals. The assessment result helps in determining the success of the task execution.

7.3 Keep/Improve/Decline

Makes a decision (keep the current approach, improve it, or decline it) based on the assessment result. This decision guides further actions in the task process.

7.4 Duplicate Removal

Removes duplicate elements from the data. This helps in keeping the data clean and efficient for further processing.

7.5 Memory/New Prompt

Stores the insights gained during the task execution and generates a new prompt based on the new task information. The new prompt can be used to adjust or continue the task.

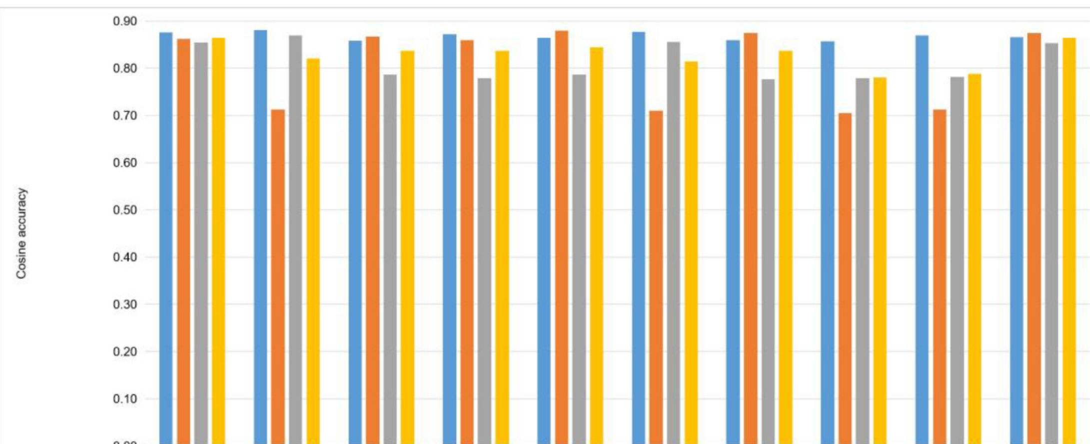
7.6 Update Current Task Understanding

Re-evaluates the task information using the feedback received. It returns the updated task information, ensuring that the agent has an accurate understanding of the task as it evolves.

Experimental results and discussion

The experiment employs the aforementioned SAA to validate its performance in HSD simulation scenarios for supporting wellbore downhole operation, using downhole operational data within years 2020–2024 from one of major shale oil producer in western China—including normal operation, pump leakage, rod fracture, and wax deposition as well as corresponding maintenance logs—as stimulus inputs. A database comprising publicly accessible research papers and global patent repositories serves as the knowledge corpus, with wellbore maintenance strategy formulation and performance evaluation as case studies.Intermediatestate outputs generated via latent space representations synthesise predictive information by integrating knowledge from normal and fault extremes through generative agents. However, as these outputs lack of empirical calibration, a control group was established, consisting of 10 volunteers from wellbore engineering and 5 volunteers from financial disciplines, all with over 5 years of professional experience. Ten unresolved intermediatestate problems were formulated by this volunteering group, each accompanied by a reference answer (≤300 words) for agent validation.To minimise subjective biases in evaluating results generated by the proposed method, the study adopts the average cosine similarity across three models—DeepSeek R1, DeepSeekV3, and ERNIE 8K—as the final metric.With agents clusters allied with foundation models, is to reason the potential results, including workflow planning, tool using, decision support, work flow executing and post evaluation were generated by thought chain based prompting (Deepseek. 2025), and attention based prediction, as a 'black-box' prediction, the confident interval was performed by foundation model and agents, which could be influenced by HIL interactions as optional process, in this study, the autonomous setting of confident interval reply by agents were varied between 0.7-0.9, which believes as the suitable range for shale-oil wellbore downhole operation case.

Analysis of the generated results (Fig. 3) indicates that the proposed framework demonstrates robust performance across decisionmaking task chains, including technical problem generation, task decomposition, workflow generation, code generation, functional diagram simulation, selfoptimisation, solution formulation and recommendation, material planning, and financial evaluation. The overall cosine similarity scores range from 0.78 to 0.86. By comparing the selected foundational models, DeepSeek R1 outperforms ERNIE 8K, as well as the smaller-scale DeepSeekV3, with all cosine similarity values exceeding 0.8. This further underscores that the deep reasoningchains facilitated by the SAA better harness the potential of application in complex task scenarios such as wellbore downhole intelligent maintenance policy decision supporting.



	Idea generation	Task decomposition	Workflow generation	Code generation	Power diagram analysis & classification	Self-refinement	Solution generation	Solution recommendation	Material planning policies	Financial calculation
Deepseek R1	0.88	0.88	0.86	0.87	0.87	0.88	0.86	0.86	0.87	0.87
DeepSeekV3	0.86	0.71	0.87	0.86	0.88	0.71	0.87	0.71	0.71	0.88
ERNIE 8k	0.85	0.87	0.79	0.78	0.79	0.86	0.78	0.78	0.78	0.85
Overall accuracy	0.86	0.82	0.84	0.84	0.84	0.81	0.84	0.78	0.79	0.86

Figure 3—Experimental results provided by using the proposed method

In specific sub-scenarios, such as solution generation, solution recommendation, and material planning, the average similarity scores are slightly lower than other contexts but remain above 0.75, still indicating high consistency. These findings suggest that integrating generative AI with proposed swarm agent based approach, augmented by external knowledge bases or online queries, enables autonomous decision support and generative digital twin based simulations in latent spaces without requiring instruction fine-tuning. By specifying prompt templates, the system elicits generative agent reasoning, producing auxiliary decisionmaking data and facilitating self-improvement through iterative refinement.For systems characterised by high reliability and infrequent abnormal conditions, this approach provides effective decision support for coordinating spare parts management, production planning, and supply chain optimisation.

Building on the proposed methodology, the downhole management system (DMS)—a swarm agents based system for wellbore downhole operationsis illustrated in Fig. 4. As illustrated, when users input ambiguous natural language queries via the interactive dialogue interface, the system autonomously executes fullcycle wellbore operations, including functional diagram generation, historical data analysis, fault evolution simulation, and well maintenance + financial evaluation, leveraging the SAA.

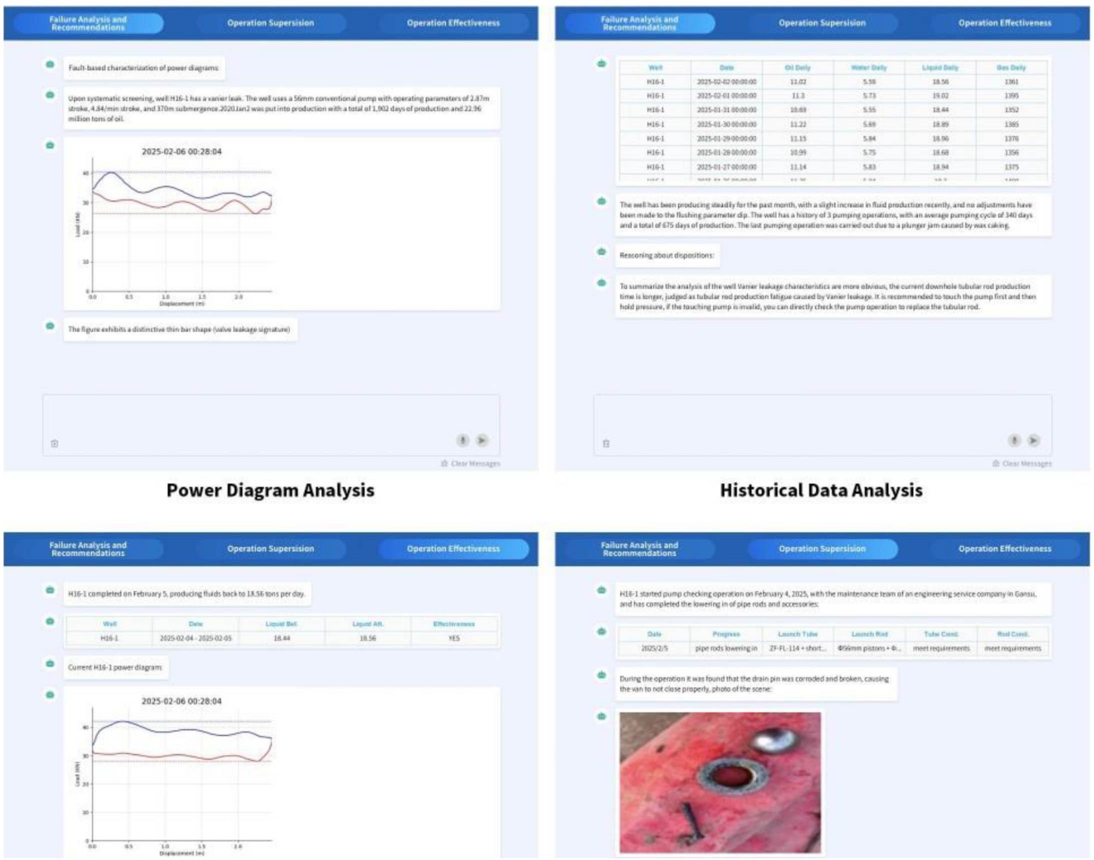


Figure 4—Swarm agent based wellbore downhole operational supporting system- an instance

Conclusion and discussion

The manuscript proposed a swarm generative agent based approach to deal with complex, high-value drilling assets operational and production management problems, enabling users to simulate, emulate, and infer prior to failuremaintenance scenarios for highvalue asset management such as wellbore. Leveraging the SAA with HSD simulation, a HIL system is constructed, encompassing hyperautomation of full process including problem identification, solution generation, data synthesis, and decision recommendation, thereby establishing a HSD framework that supports the evolution of equipment performance under abnormal conditions. Building on this HSD, autonomous model optimisation of SAA derived evolutionary data is further realised, endowing the system with self refinery capabilities. These outcomes provide a potential auxiliary decision support framework for refined, fullvaluechain decision-making and business-financial synergistic optimisation in high-reliability, high-availability equipment contexts. Furthermore, standardising this research framework could facilitate the broader application of latent spaceassisted decision-making and generative virtual-real mirroring technologies across diverse cross value chain planning management scenarios upon wellbore downhole operation as well as other complicated drilling process and application scenes.

To compare with classical methods, the swarm agent based solutions significantly reduce the time cost of wellbore operation maintenance strategies making, with the typical decision without simulation, to instead 24-48 hours in the former, the time spend on decision making based on swarm agent solution is reduced to <1.5 hours for each case within above mentioned accuracy, whereas for complex scenarios with simulation, with autonomous workflow been introduced to call various academic tools, generated codes and executed by agents, the time cost reduced to 0.5-1 day per case to instead 72-120 hours in the former, all with system and necessary data integration of CNPC's A2, A5 systems, as well as other professional industrial software. Moreover, since the improved standard framework regarding to complex wellbore downhole operation task has been developed, with the similar maintenance processes formed to wellbore downhole managements under different geo-conditions, it is worth to believe that the genAI based hyper-automation technology could be replicated to other complex process of wellbore downhole operation scenes. Additionally, similar prior success cases provided by the same team under the era of 'prior-version of AIGC agents', but in other upstream scenarios (Chen et al. 2024; Chen et al. 2025), could be another verification for scalability of our continuous growing swarm agent based approaches.

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References

Chen, H. Zhao, Q. et al 2024. Application of Self-supervised Autonomous Agent Framework for Digital Transformation of Elder Well Potential Discovery. Paper Presented at the SPE Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 4 - 7 November. SPE-223028-MS. https://doi.org/10.2118/223028-MS.

Chen, H., Zhao, Q., Lu, W., et al 2025. Application of Self-Supervised Autonomous Agent Framework With Growing Scheme for Digital Transformation of Elder/Existed Well Potentials Discovery. Paper Presented at the International Petroleum Technology Conference, Kuala Lumpur, Malaysia, 18 - 20 February. IPTC-24776-MS. https://10.2523/IPTC-24776-MS.

DeepSeek. 2025, DeepSeekR1/R2/V3, https://www.deepseek.com/ (accessed 1 June 2025).

Edge, D., Trinh, H., Cheng, N., et al 2024. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. arXiv preprint, arXiv:2404.16130. https://doi.org/10.48550/arXiv.2404.16130.

Feng, K., Pu, K., Latzke, M. et al 2025. Cocoa: Co-Planning and Co-Execution with AI Agents. arXiv:2412.10999v3. https://doi.org/10.48550/arXiv.2412.10999.

OpenAI. 2025. Openai_GPT_Products, https://openai.com/ (accessed 28 August 2025).

Ocker, F., Menzel, S., Sadik, A., et al 2025. From Idea to CAD: A Language Model-Driven Multi-Agent System for Collaborative Design. arXiv:2503.04417v1. https://doi.org/10.48550/arXiv.2503.04417.

Ren, L., H. Wang, Y. Wang et al 2025. Foundation Mmodels for The Process Industry: Challenges and Opportunites, Engineering, https://doi.org/10.1016/j.eng.2025.03.023 (In press).

Schmidgall, S., Moor, M., 2025. AgentRxiv: Towards Collaborative Autonomous Research. arXiv:2503.18102v1. https://doi.org/10.48550/arXiv.2503.18102.

Wang, R., Kuru, E., Zhao, C., et al 2024. Assessment of Wellbore Integrity Failure Risk and Hazardous Zones in Depleted Reservoirs Underground Gas Storage During the Operation Processes. *Environ. Sci. Pollut. Res.*, **31**, 2079–2089. https://doi.org/10.1007/s11356-023-31297-8.

Wang, F., Wang, Y., et al 2024. Optimizing Brand Loyalty Through User-Centric Product Package Design: A Study of User Experience in Dairy Industry. *Heliyon*, **10**, e25484. https://doi.org/10.1016/j.heliyon.2024.e25484

Xu, B., Xiao, L. 2024. Comparison of Well Logging Formation Evaluation Using Serial and Parallel Multi-task Learning Networks. *J Geophys-CH*, **67**(4), 1613–1626. https://10.6038/cjg2023R0584 (In Chinese).

Xiong, W., Xiao, L., et al 2024. Automatic Depth Matching Method of Well Log Based on Deep Reinforcement Learning. *Petrol Explor Dev*, **51**(3), 634–646. https://doi.org/10.1016/S1876-3804(24)60493-3.